

Sampling and Quantization of Signals

Lecture Deck: EE501 Digital Signal Processing

Unit 2: Analog-to-Digital Conversion & Multirate Systems

Outline:

- Nyquist-Shannon Sampling Criterion
- Aliasing and Anti-Aliasing Pre-filters
- Ideal Sinc Interpolation vs D/A Conversion
- Uniform & Non-uniform Quantization (SQNR)
- Oversampling and Multirate DSP

The Nyquist-Shannon Sampling Theorem

Fundamental Theorem of Information Conversion:

A continuous-time bandlimited signal $x(t)$ with maximum frequency component $F_{\max}$ (or $\Omega_{\max} = 2\pi F_{\max}$) can be completely and uniquely recovered from its discrete samples $x[n] = x(nT_s)$ if and only if:

$$F_s \geq 2 * F_{\max}$$

Key Definitions:

- $F_s = 1 / T_s$: Sampling frequency (samples/second or Hz)
- Nyquist Rate: $2 * F_{\max}$ (the minimum theoretical sampling rate)
- Nyquist / Folding Frequency: $F_N = F_s / 2$

Aliasing Phenomenon and Spectral Folding

What happens when $F_s < 2 * F_{max}$?

In the frequency domain, sampling causes periodic replication of the continuous spectrum $X(F)$:

$$X_s(F) = (1 / T_s) * \sum_{k=-\infty}^{\infty} X(F - k * F_s)$$

When $F_s < 2 * F_{max}$, adjacent spectral replicas overlap. High frequencies above $F_s/2$ fold back into the baseband, irreversibly corrupting the signal. Once aliasing occurs, no digital filter can separate the folded components.

Anti-Aliasing Analog Pre-Filter Design

Preventing Aliasing at the System Input:

- An analog lowpass filter MUST precede the A/D converter.
- Passband: $0 \leq F \leq F_c$, passing desired signal bandwidth with minimal attenuation.
- Stopband: $F \geq F_s / 2$, attenuating components above the folding frequency to below ADC noise floor.
- Filter Selection: Butterworth for flat passband; Elliptic for steep transition roll-off.

Signal Reconstruction: Ideal Sinc Interpolation

Reconstruction of Continuous Signals from Samples:

Under the Nyquist condition, the original continuous signal $x(t)$ is recovered through ideal lowpass filtering:

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] * \text{sinc}((t - nT_s) / T_s)$$

where $\text{sinc}(u) = \sin(\pi * u) / (\pi * u)$.

Properties:

- At $t = kT_s$, $\text{sinc}(0) = 1$ and $\text{sinc}(m) = 0$ for all $m \neq 0$, ensuring exact sample interpolation.
- Sinc filter is non-causal with infinite impulse response, requiring practical D/A approximations.

Zero-Order Hold (ZOH) Reconstruction

Practical D/A Converters: Zero-Order Hold:

- ZOH maintains sample value $x[n]$ constant for the entire period T_s : $h_{\text{zoh}}(t) = 1$ for $0 \leq t < T_s$.
- Frequency Response: $H_{\text{zoh}}(f) = T_s \cdot \text{sinc}(f \cdot T_s) \cdot e^{-j\pi f T_s}$.
- Distortion: Droop near the band edge $f = F_s/2$ of approximately 3.92 dB.
- Correction: Post-reconstruction analog smoothing filter or digital pre-equalizer.

First-Order Hold (FOH) and Post-Filtering

First-Order Hold (Linear Interpolation):

- Connects adjacent samples with straight line segments.
- Impulse response: triangular pulse of duration $2T_s$.
- Frequency response: $|H_{\text{foh}}(f)| = T_s * \text{sinc}^2(f * T_s)$.
- Provides steeper high-frequency attenuation than ZOH (-40 dB/decade vs -20 dB/decade).

Uniform Amplitude Quantization

Quantization Fundamentals:

Maps continuous amplitude $x[n]$ into a discrete set of 2^B digital representation levels.

Parameters:

- Full-scale range: $R = V_{\max} - V_{\min}$
- Number of bits: B
- Quantization step size: $\Delta = (V_{\max} - V_{\min}) / 2^B$
- Quantizer types: Midtread (zero is a representation level) and Midrise (zero is a decision threshold).

Quantization Error and Noise Modeling

Statistical Model of Quantization Error $e[n] = x_q[n] - x[n]$:

Assumptions for small Δ and complex signals:

1. $e[n]$ is uniformly distributed over $[-\Delta/2, \Delta/2]$.
2. $e[n]$ is a stationary white noise process uncorrelated with $x[n]$.

Noise Statistics:

- Mean: $\mu_e = E\{e[n]\} = 0$
- Quantization Noise Variance: $\sigma_q^2 = \int_{-\Delta/2}^{\Delta/2} e^2 * (1/\Delta) de = \Delta^2 / 12$.

Signal-to-Quantization-Noise Ratio (SQNR)

Theoretical SQNR for Sinusoidal Signals:

Let input be a full-scale sinusoid $x(t) = (V_{\max} - V_{\min})/2 * \sin(\omega t)$ with peak-to-peak range $2^B * \Delta$.

Signal Power: $P_s = (2^B * \Delta / 2)^2 / 2 = 2^{2B} * \Delta^2 / 8$.

Noise Power: $P_n = \sigma_q^2 = \Delta^2 / 12$.

SQNR Ratio = $P_s / P_n = (3/2) * 2^{2B}$.

In Decibels:

$SQNR \text{ (dB)} = 10 * \log_{10}(3/2) + 20 * B * \log_{10}(2) = 6.02 * B + 1.76 \text{ dB}$.

Takeaway: Every additional bit added to the ADC increases SQNR by ~6 dB.

Non-Uniform Quantization & Companding

Speech and Audio Quantization:

- Problem: Small amplitude speech signals suffer poor SQNR with uniform quantization.
- Solution: Companding (Compressing before uniform quantization, Expanding after D/A).
- Standards:
 - * μ -Law Companding (North America & Japan): $F(x) = \text{sgn}(x) * \ln(1 + \mu * |x|) / \ln(1 + \mu)$, $\mu = 255$.
 - * A-Law Companding (Europe & International): $A = 87.6$.

Dynamic Range Considerations in Hardware

Dynamic Range (DR) in Digital Signal Processors:

- 16-bit fixed point: ~96 dB dynamic range.
- 24-bit audio: ~144 dB dynamic range (exceeds human hearing threshold).
- 32-bit floating point: IEEE-754 format provides > 1500 dB dynamic range, virtually eliminating internal overflow.

Oversampling and Noise Shaping (Sigma-Delta ADC)

High-Resolution Data Conversion Principles:

- Oversampling: Sampling at $F_s \gg 2 * F_{\max}$ spreads quantization noise power $\Delta^2/12$ over a much wider bandwidth.
- Oversampling Ratio (OSR): $OSR = F_s / (2 * F_{\max})$.
- Noise Shaping: Feedback loop filters quantization noise, pushing noise power from baseband to high out-of-band frequencies.
- Digital Decimation Filter: Removes out-of-band noise and reduces sample rate to Nyquist rate.

Multirate DSP: Decimation (Downsampling)

Decimation by Integer Factor M:

- Downsampler keeps every M-th sample: $y[m] = x[m * M]$.
- Spectrum: $Y(e^{j\omega}) = (1 / M) * \sum_{k=0}^{M-1} X(e^{j(\omega - 2\pi k)/M})$.
- Anti-Aliasing Decimation Filter: To avoid aliasing during downsampling, signal MUST be filtered by digital lowpass filter with cutoff $\omega_c = \pi / M$ prior to sample removal.

Multirate DSP: Interpolation (Upsampling)

Interpolation by Integer Factor L:

- Step 1: Zero-stuffing: Inserts $L - 1$ zeros between consecutive input samples:
 $x_u[n] = x[n / L]$ for n a multiple of L , and 0 otherwise.
- Step 2: Anti-Imaging Lowpass Filter: Cutoff frequency $\omega_c = \pi / L$ and gain L to remove spectral images at multiples of $2\pi/L$.

Polyphase Filter Structures

Efficient Realization of Multirate Filters:

- Standard FIR filtering before decimation computes samples that are subsequently discarded.
- Polyphase Decomposition decomposes transfer function into M branches: $H(z) = \sum_{k=0}^{M-1} z^{-k} * E_k(z^M)$.
- Noble Identities allow moving downsamplers and upsamplers past filter blocks, running all filtering operations at the lowest sample rate.

Quadrature Mirror Filter (QMF) Banks

Subband Coding and Frequency Splitting:

- Two-channel QMF bank splits signal into low-frequency and high-frequency subbands.
- Analysis filters: $H_0(z)$ (lowpass), $H_1(z) = H_0(-z)$ (highpass mirror).
- Synthesis filters: $G_0(z) = 2H_0(z)$, $G_1(z) = -2H_1(z)$.
- Perfect Reconstruction (PR) cancels aliasing between subband channels.

Review and Exam Highlights

Key Formula Summary for Exam Night:

1. Sampling Criterion: $F_s \geq 2 * F_{\max}$
2. Folding Frequency: $F_N = F_s / 2$
3. Quantization Noise Variance: $\sigma_q^2 = \Delta^2 / 12$
4. Sinusoidal SQNR Rule: $SQNR = 6.02 * B + 1.76 \text{ dB}$
5. Decimation Cutoff: $\omega_c = \pi / M$
6. Interpolation Cutoff: $\omega_c = \pi / L$